

Chapter no 6:-

Science

Definition of science:-

- **Philosophy of science:-**

"philosophy of science is the branch of philosophy that explores the scientific inquiry including the scientific methods, the nature of scientific knowledge and the relationship between science and the philosophy"

Questions:-

What is Science?

What is the role of science?

• Definition of science:-

Science is a systematic way of searching for truths and understanding the natural world through observations, experimentation and evidence.

Science involves:

- Asking questions
- making observations
- testing ideas
- Finding answers.

⇒ Science help us to understand the world around us, the things around us, how these things work and why things happened

Branches:-

There are so many branches of science and some of them are:-

Natural science:

It's deals with the study of natural world

e.g. Biology, Chemistry, Physics.

Formal Science:

It's deals with the study of logic, mathematics, and statistics.

Social Science:

"It's deal with the study of human behaviours and societies."

e.g. Psychology, sociology, anthropology.

Applied Science:-

"It's deal with the study of practical applications and of scientific knowledge"

e.g. Engineering, medicine e.t.c

=> Karl Popper's theory of falsification:-

Karl Popper, a prominent 20th-century philosopher of Science introduced the theory of falsification. He defined it as:

"It is a philosophical approach to scientific inquiry that emphasizes the role of criticism and falsification in the development of scientific knowledge."

Method:-

Formulate a Hypothesis:-

Propose

a scientific theory

Test the hypothesis:-

Designs

experiments to test the theory

Attempt to falsify:-

Try to

prove the theory false through empirical evidence

Evaluate result:-

If the theory is falsified then reject and modify it.

Examples:-

Formulate a hypothesis:-

white
All swans are

Test the hypothesis:-

observe swans
in different parts of world

Attempt to falsify:-

look for

a non-white swan

• **Discovery of black swan:-**

observation

of black swan in Australia
falsifies the hypothesis.

Evaluate result:-

Reject the

hypothesis "All swans are white"
and formulate a new one

"Not all swans are white"

Implications of falsificationism:

• **Scientific Rigor:-**

Falsification

encourages scientists if there's

old hypothesis turns out to be false, they should not lose heart, but rather formulate a new hypothesis. So, it encourages them.

Progress of science:-

By focusing on falsifiability, science made more progress and advance through the elimination of false theories and improvement of knowledge over time.

Conclusion:-

Karl Popper theory of falsification concluded that we can't know for sure if the theory is true, but we can prove a theory is false with just one contradicting example. This helps us to improve and refine our theories over time.

⇒ Thomas Kuhn's theory of Scientific revolution:-

Thomas Kuhn's theory of scientific revolutions, presented in his influential book "The Structure of Scientific revolutions" (1962). Kuhn argued that scientific progress is not linear but occurs through a series of transformative shifts in paradigms.

Key elements:-

⇒ Paradigms and Normal Sciences:-

Kuhn introduced the concept of a paradigm, which he defined as the accepted framework of theories and methods that guide scientific research within a particular field.

Paradigm:

A dominant model that shapes how scientists understand and investigate phenomena.

e.g. Newtonian mechanics is a paradigm.

=> **Characteristics:-**

Anomalies and Crisis:-

over time, scientific research necessarily encounters anomalies - observations or experimental results that cannot be explained by the current paradigm. As anomalies accumulate, they may begin to undermine confidence in the paradigm. When the paradigm can no longer explain key phenomena, a crisis occurs.

Incommensurability:-

Kuhn argued that paradigms are incommensurable, meaning they are based on fundamentally different assumptions and cannot be directly compared.

Conclusion:-

Kuhn concluded science progresses through big revolution rather than small steps. He also concluded that new ideas replace old ones, changing our understanding about the world. This theory helps science move forward and made progress.

⇒ Problem of induction:-

The problem of induction is defined as
"It is a philosophical problem that questions the rationality

of Prediction and unobserved things based on previous observations."

Method:-

1) Pragmatic Approach:-

Assume induction works because it has been successful in the past

2) Probabilistic Approach:-

Use probability theory to quantify the likelihood of future events based on past observations-

3) Falsificationism:-

Test and eliminate false Hypothesis-

Examples:-

⇒ Sun-rise Problem:-

The sun has rise every morning so far, doesn't

mean it will ^{definitely} rise tomorrow -
Rotten apple Problem:-

you eat 10 apples none of them are rotten. You assume all apples are good but the next one apple you eat might be rotten -

Conclusion:-

These two examples concludes that how induction can lead to incorrect assumption - Just because something happens a certain way multiple times doesn't mean it's will always happen that way -